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A lightweight multiscale attention network for 3D tumor segmentation in PET images

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Accurate segmentation of tumor PET images has a significance in disease monitoring and clinical treatment. However, the task of accurately segmenting PET images remains challenging due to insufficient access to spatial features of PET images. Therefore, we propose a multiscale attention-guided feature mechanism for 3D tumor segmentation (MA3DSeg) to achieve accurate segmentation of tumor PET images. Specifically, we first utilize a 3D-DX block to construct a lightweight encoder to minimise the redundant information generated across channel features and reduce the model complexity. Additionally, we employ multiscale attention-guided feature enhancement (MAFE) module for 3D networks to fuse channel features and spatial features to fully acquire spatial feature analysis of tumors. Furthermore, we propose explicit and implicit difference information in interaction boundary semantic (IBS) module to enhance the representation of segmentation edges. We conduct rigorous experiments on three publicly available datasets ECPC-IDS, Hecktor 2022 and AutoPET, which demonstrated superior performance than other competitive networks. MA3DSeg significantly improves Dice by 0.5% on Hecktor 2022 and RVD by 2.1% on ECPC-IDS compared with 3D UX-Net. The experimental results show that MA3DSeg achieves excellent tumor segmentation performance and generalization ability.

Keywords Tumor segmentation, Lightweight network, Multiscale attention-guided, PET images

Oncological diseases pose a major threat to human health, such as lymphomas, uterine fibroids and soft tissue tumors. Tumors show remarkable diversity in morphology, size and location, which adds to the complexity of clinical diagnosis. Therefore, automatic and accurate segmentation of tumors in Positron Emission Tomography (PET) and Computed Tomography (CT) images is an important guide for physicians clinical diagnosis¹. PET imaging requires the injection of a fluorodeoxyglucose (18F-FDG) as a tracer, and depending on the metabolic characteristics of the tumor². However, in addition to the tumor, the body's metabolic organs also absorb large amounts of 18F-FDG, leading to confusion in differentiating between lesions.

The continuous development of deep learning has led to tremendous progress in PET image tumor segmentation. PET images are volumetric data types. It has been investigated how to segment volumetric data using 3D convolutional neural networks (CNNs) to expand the receptive field. Convolutional operations focus on local context, and lack attention to global context, leading to poor performance in processing texture-related targets.

To compensate for this, researchers have begun to explore the application of Transformer technology to the segmentation of medical images. In the field of 3D image segmentation, the UNETR³ and the Swin UNETR⁴ architectures demonstrate the capability of Transformer in extracting high-quality feature representations and exploiting global information. To improve segmentation accuracy by directly processing 3D along with patch and utilizing the Swin Transformer⁵ encoder. Nevertheless, Transformer has limitations in insufficient utilization of local features: the global attention mechanism of Transformer weakens the local spatial features. 3D medical scanning images are composed of multiple slices, to mine the associations of neighboring slices of tumor images, some researchers have used Context Residual Networks (ConResNet)⁶ to introduce a new contextual residual decoder in the coder-decoder structure. Zhou et al.⁷ proposed feature pyramids to fuse multiscale contextual information by using dilation convolution. DeepMedic⁸ inputs multi-scale features from two channels to take into account both local and global information. nnUNet⁹ introduces optimize preprocessing, training, inference, and post-processing with minor modifications on the encoder-decoder. The traditional Transformers typically

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process each slice independently or adopt a simple stacking approach, failing to fully leverage the contextual information between slices.

With the in-depth study of medical image segmentation models utilizing the attention mechanism modules, spatial features are fully exploited, such as spatial attention¹⁰, channel attention¹¹ and pixel-level attention¹². These modules have been shown to play an effective role in medical image segmentation networks^{13,14}. In the combined CNN(Convolutional Neural Network) and Transformer network, the high-resolution stage contains detailed information such as the texture and edges of the segmented target, while the low-resolution stage contains the attributes of the target¹⁵. Therefore, combining high-resolution and low-resolution feature maps can also better present the different size and morphological features of tumors^{16,17}.

Based on the methods mentioned above, the limitations of 3D image segmentation include high computational cost and large memory consumption, which hinder the effective capture of the spatial relationships between slices. As a result, the feature representation of information within the slices is limited, making it difficult to capture the internal slice information of the 3D volume.

To address the above issues, we propose a novel network enhanced with multiscale attention-guided feature mechanism for 3D tumor segmentation (MA3DSeg) with fewer parameters and less consumption. MA3DSeg has been carefully designed to prioritize initial level spatial feature analysis before shifting the focus to deeper level channel-based features that ensures a comprehensive feature extraction process.

Inspired by 3D UX-Net¹⁸, we use depthwise separable convolution in the encoder, which significantly reduces the number of parameters, thus building lightweight network models. In decoder, multiscale attention-guided feature enhancement (MAFE) is introduced to capture a wide range of spatial information and augment the tumor features in the spatial dimension, and extract the fusion features from the context using global spatial and channel attention mechanisms. In addition, we propose the interaction boundary semantic(IFS) module that focuses on the edge segmentation region to improve the representation of edge details using explicit and implicit difference information. The contributions of our work can be summarized as follows:

1. We propose a 3D tumor segmentation network MA3DSeg, which employs an innovative 3D-DX block in its encoder. This block effectively combines depthwise separable convolution with plain design for feature extraction. The encoder combines the computational efficiency of CNNs and the capability of hierarchical transformers significantly reduces the required Params and FLOPs.
2. We develop MAFE to efficiently fuse channel and spatial feature interactions of 3D image slices. This module can facilitate image connectivity across slices, enrich volumetric features through spatial context and enhance tissue classification and organ edge depiction.
3. To address the issue of inaccurate edge segmentation, we design IFS module based on diffusion theory. It facilitates the diffusion process through deep semantic information, which is used to refine the features of the blurred boundary region.

Related work

Tumor segmentation

Medical oncology imaging is PET and CT 3D images, so existing tumor segmentation methods focus on the challenges of segmentation arising from the imaging characteristics of the lesion region. Zou et al.¹⁹ proposed a two-branch tumor segmentation network using multipath downsampling to mitigate the effect of normal organ tissue features to improve segmentation accuracy. Huang et al.²⁰ proposed ISA-Net to improve spatial attention, sampling multi-scale convolutional operations to obtain accurate positional information of tumors. Boundary segmentation of tumors has also made great progress, and existing methods mainly extract boundary features by enhancing the loss function^{21,22}. However, how to overcome the continuous edge features of 3D slice images still needs to be urgently addressed to achieve more robust tumor segmentation.

Attentional mechanism

In pixel-level vision tasks, the use of attentional mechanisms can be more targeted in extracting features of the target, leading to an increase in the accuracy of the task in question. Chen et al.²³ proposed multi-attention fusion, which activating attentional mechanisms using cross-resolution fusion and scale-awareness, thus enhancing the coding ability of the model. However, the current approach focuses on the weight share of features on spatial attention and channel attention and does not comprehensively consider the global interactions between voxels. MSRA-Net²⁴ enhances spatial domain, frequency domain and channel attention through multiple receptive fields for localization of localized lesions. In addition, Attention U-Net²⁵ proposes an attention gate that uses soft attention to realign the output features at each resolution of the encoder. CANet²⁶ network employs multi-resolution input versions and multi-scale convolutional units to capture a large number of complementary features at different layers, which helps in learning local detail features of various scales.

PET segmentation

Due to the importance of accurate segmentation of PET images for clinical treatment, the field of tumor segmentation based on deep learning has progressed significantly. Most of the current results focus on the study of extracting 3D slice image features with multiple scales, as well as edge segmentation. Kang et al.²⁷ constructed a feature extractor that extracts interpretable masks from the input, which helps to enhance feature extraction. Niu et al.²⁸ introduced a simultaneous lung cancer lesion segmentation network, which accurately describes the edge region by dynamically correcting the weights and thresholds. Diao et al.²⁹ constructed an evidence fusion network for PET-CT tumor segmentation based on uncertainty theory and achieved good segmentation results. Liu et al.³⁰ proposed a method to introduce deep convolutional blocks into ConvNeXt, combining spatial and

channel attention, to generate 3D medical image feature information containing multiple feature dimensions to improve segmentation performance.

Methods

Network architecture

The overview of MA3DSeg architecture is illustrated in Fig. 1. In encoder, we consider taking advantage of the respective strengths of Swin Transformer and ConvNet to further complement each other. To this end, we develop a convolutional model named 3D-DX block which is inspired by ConvNeXt. And we have redesigned the hierarchical transformers based on 3D-DX block to further improve the usefulness of volumetric segmentation in 3D medical image datasets. The decoder goes through four upsampling stages, each of which seamlessly integrates three modules. MAFE module is designed to enhance the attention mechanism of the 3D network from spatial and channel dimensions. To achieve accurate segmentation of PET image edges, IBS is proposed to effective extraction of edge features based on diffusion theory to enhance lesion edge segmentation. Furthermore, Res-Block is used to assist IBS for tumor boundary segmentation.

3D-DX block

We propose 3D-DX block which further leverage convolution modules to enable the capabilities of hierarchical transformers. The 3D-DX block introduces several ConvNeXt priors, which complement the strength of Transformer to achieve lightweight and efficient 3D medical segmentation networks. It uses deep convolution with fewer model parameters to allow the network to learn how to capture a larger receptive field.

Specifically, 3D-DX block consists of double Depthwise Convolution, including Depthwise Separable Convolution (DSC) and Depthwise Convolutional Scaling (DCS), as shown in Fig. 2. We acquire volumetric image features using depth-wise convolution with large kernel (LK) size to achieve a larger global receptive field. DSC employs LK with $7 \times 7 \times 7$ to extract initial feature for more global features, utilises LK-based deep convolution as a feature extraction backbone. It enhances the capture of spatial information through the large kernel convolution, which acts as the backbone of feature extraction to increase the receptive field, and enriches the features by extending the independent channels. DCS with $1 \times 1 \times 1$ kernel size to independent linear scaling for each channel feature. Enriching the feature representation by scaling and compressing each channel independently, the redundant information generated by the cross-channel context can be minimized. Finally, using the same DCS reduces the number of parameters required for convolutional computation and reduces the complexity of the model.

3D-DX based encoder

The encoder of MA3DSeg consists of four stages and constructed based on 3D-DX block at each stage. Instead of the traditional flattening the patches operation, we use LK convolutional layer to compute the partitioned feature maps to obtain a 3D image volumes of $X_i \in \mathbb{R}^{\frac{H}{2} \times \frac{W}{2} \times \frac{D}{2} \times 48}$. To maintain the computational local self-attention advantage, and reduce the model computational parameters, we replace the shifted window mechanism by uniformly chunking the feature map using DSC with kernel size of $7 \times 7 \times 7$ with padding of 3. DCS is used as the main scaling operator of DSC, which is designed to achieve an optimal balance between ViTs and large convolutional kernels, and efficiently acquire the features in the global receptive field. The encoder outputs of each layer are defined as follows:

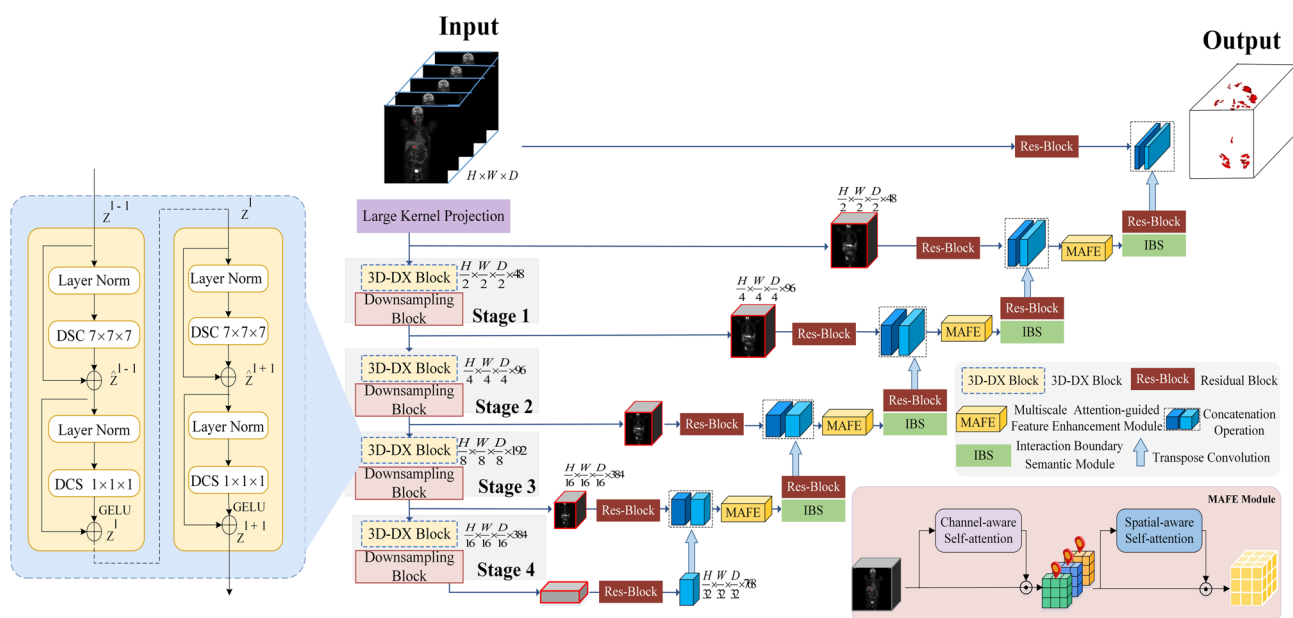


Figure 1. The overview of MA3DSeg architecture.

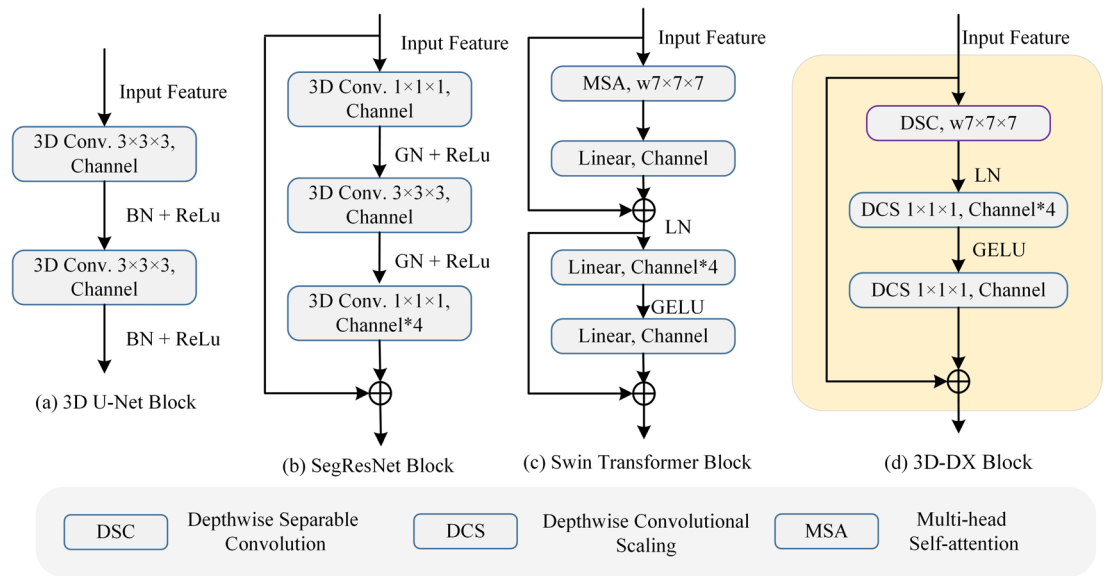


Figure 2. This figure compares our proposed block designed with double depthwise convolution.

$$\hat{z}^l = DSC \left(LN \left(z^{l-1} \right) \right) + z^{l-1} \quad (1)$$

$$z^l = DCS \left(LN \left(\hat{z}^l \right) \right) + \hat{z}^l \quad (2)$$

$$\hat{z}^{l+1} = DSC \left(LN \left(z^l \right) \right) + z^l \quad (3)$$

$$z^{l+1} = DCS \left(LN \left(\hat{z}^{l+1} \right) \right) + \hat{z}^{l+1} \quad (4)$$

where $\hat{z}^l$ and $\hat{z}^{l+1}$ are the outputs from the DSC layer in different depth levels, LN denotes the layer normalization. where $\hat{z}^l$ and $\hat{z}^{l+1}$ are the outputs from the DSC layer in different depth levels, LN denotes the layer normalization.

Multiscale attention-guided feature enhancement (MAFE)

The 3D image consists of multiple slices with insufficient feature correlation between the slices and does not adequately consider the critical interactions between spatial and channel features. Therefore, we propose MAFE module, which addresses the enhancement of multiscale attentional features. MAFE sequentially consists of Channel-aware Self-attention Module (CSM) and Spatial-aware Self-attention Module (SSM).

Channel-aware self-attention module (CSM)

The CSM module combines branches and channels by concatenating three different scales of channel branches from the input feature map, which is used to efficiently capture global context features and multiscale channel dependencies as shown in Fig. 3. Firstly, each attention point is computed to derive its query vector, value vector, and global channel key matrix K^{ch} . Then, the matrix Q^{ch} consisting of the query vectors of all the attention points is subjected to a matrix K^{ch} multiplication operation to obtain the multiscale channel attention score matrix Z^{ch} . Finally, the matrix V^{ch} and matrix Z^{ch} are subjected to element-wise multiplication to enhance the multiscale channel features. We define the input feature of the CSM module as $A \in \mathbb{R}^{C \times H \times W \times D}$, the resolution of the input feature map is $C \times H \times W \times D$ ($144 \times 20 \times 20 \times 12$), and channel-aware self-attention score matrix $Z^{ch} \in \mathbb{R}^{C \times 1 \times 1 \times 1}$:

$$Q^{ch} = \sigma_q \left(W_q(A) \right) \quad (5)$$

$$K^{ch} = F_{SM} \left(\sigma_k W_k(A) \right) \quad (6)$$

$$Z^{ch} = F_{SG} \left(F_{LN} \left(W_z \left(Q^{ch} \times K^{ch} \right) \right) \right) \quad (7)$$

where σ_q, σ_k refer to different reshape operations. W_q, W_k, W_z denote convolution layers, respectively. $F_{SM}()$ denotes $F_{SM}(x) = \frac{e^x}{\sum_{j=1}^N e^{x_j}}$. $F_{LN}()$ represents layernorm operation. $F_{SG}()$ represents sigmoid activation function. The output of CSM module is the feature map after channel-aware enhancement $A^{ch} \in \mathbb{R}^{C \times H \times W \times D}$:

$$V^{ch} = W_v(A) \quad (8)$$

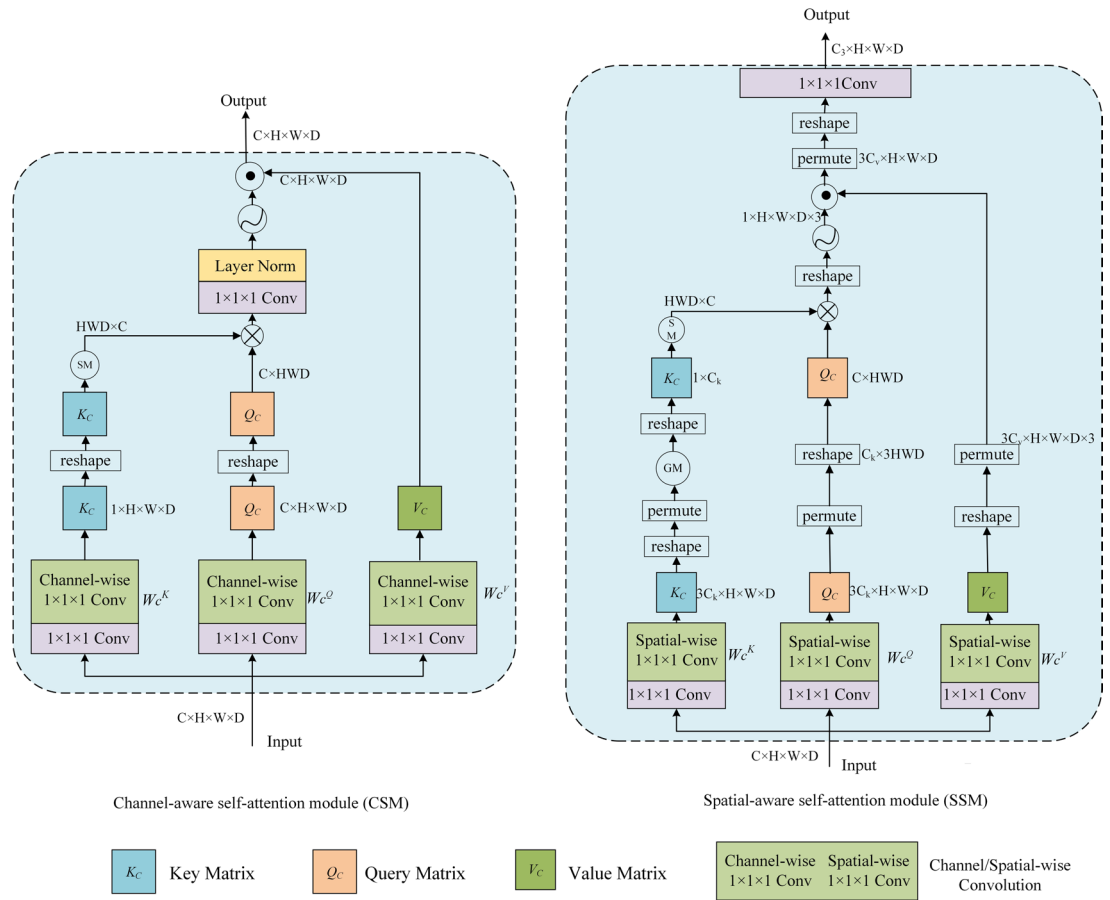


Figure 3. The Channel-aware self-attention module and spatial-aware self-attention module.

$$A^{ch} = V^{ch} \odot Z^{ch} \quad (9)$$

where W_v is convolution layer, and $\odot$ denotes the element wise product.

Spatial-aware self-attention module (SSM)

To capture the multi-scale spatial dependencies between PET image slices, SSM introduces the benefit of a mechanism that combines branch and space to enhance features more accurately while obtaining a global context. In Fig.3, we compute the query vector, the value vector, and the global spatial key matrix K^{sp} for each attention point firstly. Then, the matrix Q^{sp} consisting of the query vectors of all attention points is multiplied with this matrix K^{sp} , which in turn yields the multiscale spatial attention score matrix Z^{sp} . Finally, the matrix Z^{sp} and matrix V^{sp} are multiplied element-wise to achieve enhancement of the multiscale spatial features.

The features enhanced by the CSM module channel-aware is used as inputs to SSM $A^{ch} \in \mathbb{R}^{C \times H \times W \times D}$, spatial-aware self-attention score matrix $Z^{ch} \in \mathbb{R}^{1 \times C \times H \times W \times D \times 3}$ is defined as:

$$Q^{sp} = \sigma_q (P_q (W_q (A^{ch}))) \quad (10)$$

$$K^{sp} = F_{SM} (\sigma_k^2 (F_{GM} (P_k (\sigma_k^1 (W_k (A^{ch})))))) \quad (11)$$

$$Z^{sp} = F_{SG} (\sigma_z (K^{sp} \times Q^{sp})) \quad (12)$$

where W_q and W_k are convolution layers, respectively. $\sigma_q, \sigma_k^1, \sigma_k^2$ and σ_z are different reshape operations. P_q, P_k represent dimension permute operations. $F_{GM}()$ denotes global mean operation.

The output of the SSM module is the feature map after spatia-aware enhancement $A^{sp} \in \mathbb{R}^{C_3 \times H \times W \times D}$:

$$V^{sp} = P_v (\sigma_v (W_v (A^{ch}))) \quad (13)$$

$$A^{sp} = W_{out} \left(\sigma_{out} \left(P_{out} \left(V^{sp} \odot Z^{sp} \right) \right) \right) \quad (14)$$

where W_v, W_{out} are convolution layers. P_v, P_{out} refer to dimension permute operations. σ_v, σ_{out} denote different reshape operations.

Interaction boundary semantic (IBS) module

In PET images of tumor lesions, blurring of the edges between the lesion and normal body tissues as well as discontinuities in the lesion area are difficulties in segmentation. In Fig. 4, the IBS module based on differential operators to achieve accurate classification of uncertain edge pixels by means of learnable boundary operators. The explicit and implicit differential information of IBS greatly improve the representation of boundary volumetric classification.

Due to the variability of distribution in 3D tumor PET images in x, y and z dimensions, traditional image-based edge operators are unable to obtain edge features. Therefore, we introduce the diffusion function in the edge features P of the precedent decoder to locate the uncertain boundary with semantic information. This is done by using the depth feature ∇P difference map as the interaction semantic guidance map, while the squared difference $g(|\nabla P|^2)$ is used to fit the nonlinear features of the diffusion process ground. The differential features are calculated using a kernel applied to the feature map. This kernel calculates the difference between the current feature and its neighbors. The interaction boundary semantic mathematical formula is:

$$\hat{F}_m^{t+1} = \sum_{\tilde{m} \in \delta_m} g(|P_{\tilde{m}} - P_m|^2) \bullet (F_{\tilde{m}}^t - F_m^t) \quad (15)$$

$$F_m^{t+1} = \lambda \bullet F_m^t + \rho \bullet \hat{F}_m^{t+1} \quad (16)$$

where g is a convolutional projection function, δ_m is a local neighborhood region of size $3 \times 3 \times 3$ with m as the center point, λ and ρ are weighting coefficients. $F_{\tilde{m}}^t - F_m^t$ is the differential information of the original feature F^t at point m , represents rich boundary information, which contains irregular tumor edges in 3D slice images. Furthermore, diffusion process constraints are added to accurately predict the interclass boundaries.

Thus, the interaction semantic difference guidance $|P_{\tilde{m}} - P_m|^2$ is defined as a learnable boundary operator,

which introduced to generate refined boundary feature $\hat{F}^{t+1}$. As t increases, $|P_{\tilde{m}} - P_m|^2$ will diffuse to a stable state as a way of highlighting the boundaries between different classes and acting as a disincentive to activation in other irrelevant boundary regions. Then the original feature F^t and the enhanced boundary feature $\hat{F}^{t+1}$ are fused to get the refined feature F^{t+1} , where $\lambda = 0.6$ and $\rho = 0.4$. By this method thereby capturing the edge information in the fine-grained features.

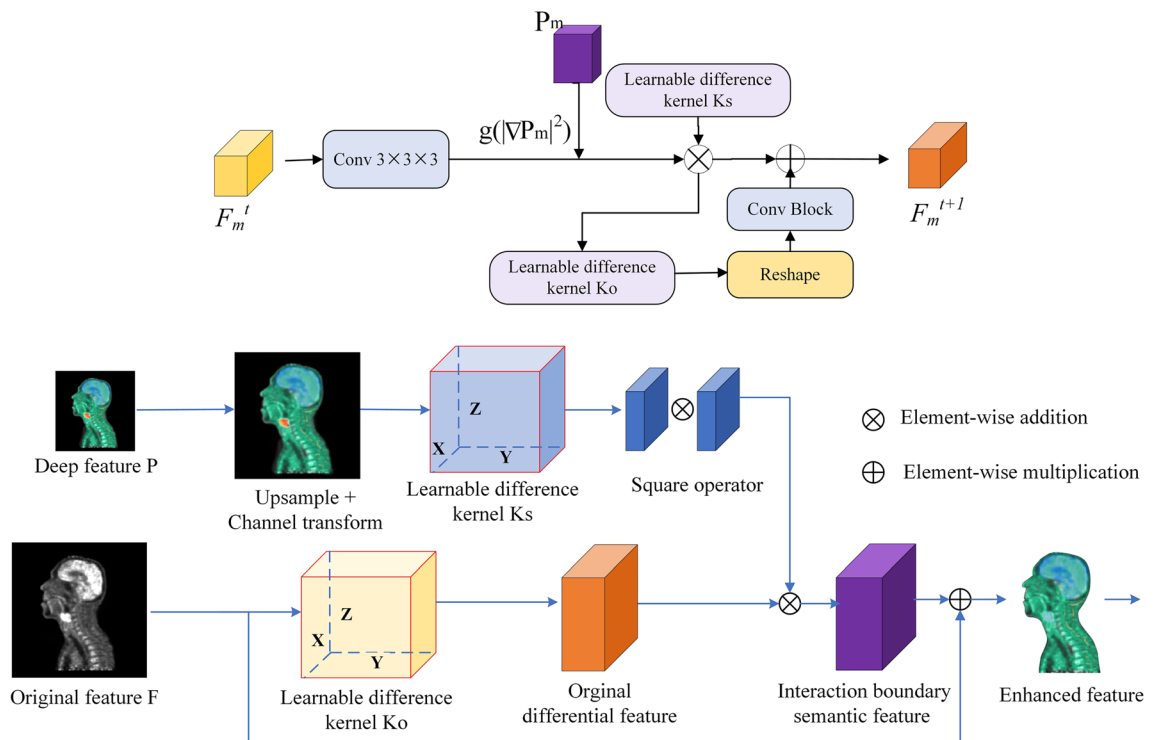


Figure 4. The illustration of the IBS module.

Loss Function

The Dice loss function is deal with difficult unbalanced categories. It is formulated as follows.

$$Dice_Loss = 1 - \frac{2 \sum_{i=1}^N \sum_{j=1}^C y_{i,j} p_{i,j} + \varepsilon}{\sum_{i=1}^N \sum_{j=1}^C (y_{i,j} + p_{i,j}) + \varepsilon} \quad (17)$$

where p_i and y_i denote the values of the prediction result and the true label at the i -th pixel point, respectively. ε is a very small value of the smoothing term that avoids division by zero. Simultaneously, we introduce the Focal loss function combination to form the overall loss function to compensate for the defect that Dice Loss will overfit when training on small targets in PET images. It is defined as:

$$Focal_Loss(p_t) = -(1 - p_t)^\gamma \log(p_t) \quad (18)$$

where p_t denotes the predicted probability of the model for the target class. γ is used to adjust the weights of the difficult and easy samples. Overall, we combine Focal and Dice loss functions for learning difficult samples as well as classification of boundary pixels, which is a good solution to the class imbalance problem of tumor images. The formulation of the proposed loss function is as follows:

$$Loss_{Function} = \alpha Dice_Loss(p_t) + \beta Focal_Loss(p_t) \quad (19)$$

where ε , β are the weighting parameters, the experimental best result parameters value were $\alpha = 0.4$ and $\beta = 0.6$.

Experiments

Datasets

In this study, we evaluate our MA3Dseg using three public datasets in Table 1. ECPC-IDS dataset³¹ is a publicly available PET/CT image dataset of endometrial cancer for evaluating semantic segmentation and high metabolic region detection. The Hecktor 2022 database³² has PET/CT scan images of 524 confirmed patients with oropharyngeal head and neck cancer. The AutoPET dataset³³ consists of 900 patients with malignant melanoma, lymphoma or lung cancer collected by FDG-PET/CT examinations at two large medical centers in Germany.

Data preprocessing

Due to the large number of 3D PET image slices, in order to eliminate invalid boundaries in the background region as well as to speed the model training, the slices were cropped to $256 \times 256 \times 256$ voxel size with the tumor in the centre and normalised preprocessing was carried out to improve the generalization ability and robustness of the model.

Evaluation metrics

To perform a comprehensive evaluation of the proposed MA3Dseg network, we choose four metrics to assess the segmentation performance. Dice is used to measure the overlap between the segmentation result and the ground truth, and Relative Volume Difference (RVD) indicates the volumetric difference in segmentation results relative to the true situation. They are formulated as:

$$Dice = \frac{2 \bullet |pred \cap true|}{|pred| + |true|} \quad (20)$$

$$RVD = \frac{|pred| - |true|}{|true|} \quad (21)$$

where $pred$ and $true$ represent the prediction region and the labelling region, respectively. HD95(Hausdorff Distance) is used to measure the segmentation accuracy of the boundary. It is formulated as:

$$HD_{95} = \max_{95\%} \left\{ \max_{g_i \in g} \min_{p_j \in p} d(g_i, p_j), \max_{p_i \in p} \min_{g_j \in g} d(p_i, g_j) \right\} \quad (22)$$

where $d(g_i, p_j)$ represents the distance from g_i to p_j . Average Symmetric Surface Distance (ASSD) calculates the average distance between the predicted boundary and the actual boundary. It is given by:

Data	Train set (slice level)	Valid set (slice level)	Total
ECPC-IDS	953	240	1193
Hecktor 2022	1408	515	1923
AutoPET	2263	437	2700

Table 1. Division of the datasets.

$$ASSD = \frac{\sum_{g \in S_g} d_{st}(g, S_p) + \sum_{p \in S_p} d_{st}(p, S_g)}{|S_g| + |S_p|} \tag{23}$$

where S_p represents the surface voxels in the set p , and $d_{st}(g, S_p)$ denotes the shortest distance from voxel g to all the voxels of set S_p .

Implementation details

All experiments are conducted in PyTorch framework empowered by NVIDIA RTX 3090 24G GPU. The training was performed employing the Adam optimizer to minimize loss function. The learning strategy for the MA3DSeg is learning rate of 1×10^{-4} and mentum of 0.9 initially. In addition, the models were trained for 200 epochs and batch size set to 12.

Experimental results

Comparison with the competitive methods

We conducted comparison experiments of MA3DSeg with the VNet³⁴, nnUNet, AttnUNet, UNETR, Swin-UNETR, 3DUX-Net, ISA-Net³⁵ and MEFN³⁶ networks on two datasets.

Quantitative Comparison: In Table 2, Dice and RVD reach 0.783 and 0.267, which are improved by 0.9% and 2.1% compared to 3D UX-Net on ECPC-IDS dataset. ASSD of MA3DSeg reaches the optimal value 2.443, which is 8.8% and 7.4% lower than 3D UX-Net and UNETR respectively, indicating that MA3DSeg achieves the highest accuracy in segmenting the boundary. The MEFN has achieved excellent performance on the HD95 metric, outperforming ISA-Net and MA3DSeg by 0.424 and 0.546 respectively. On Hecktor 2022 dataset, Dice of MA3DSeg is improved by 0.5% and HD95 is improved by 0.947 compared to 3D UX-Net, which indicates that the segmentation volume of lymphoma is closest to the label. ASSD is 0.181 lower than Swin-UNETR and 0.101 lower than UNETR, which means that MA3DSeg has a higher quality in segmenting lymphoma boundaries and achieves a better boundary matching. Therefore, MA3DSeg has the best segmentation results when quantitatively comparing the network segmentation performance of the two datasets.

We further evaluate the model parameters and computational complexity on Hecktor 2022. In MA3DSeg, we further substitute the multi-layer perceptron (MLP) in 3D-DX blocks with depthwise convolutional scaling and enhance model performances with fewer normalization and activation layers, thus the required parameters is significantly reduced. We revisit volumetric depth-wise convolutions with LK size to reduce model complexity. We further compared different model parameters and computational complexity to verify the efficiency of the MA3DSeg network as shown in Fig.5. Compared with MEFN, although their parameter scales are smaller, the proposed MA3DSeg requires fewer FLOPs, reducing Flops by 26.1% and improving the Dice coefficient by 0.5% on Hecktor 2022 dataset. Experiment results have demonstrated that our method uses the fewest parameters and lower FLOPs with an optimal balance between accuracy and complexity compared to other methods.

Qualitative Comparison: We selected three PET images of different tumor shapes from each of the ECPC-IDS and Hecktor 2022 datasets as shown in Fig.6. The third image is interfered by the background noise, some normal tissues around the tumor are classified as lesion areas in AttnUNet. Compared with UNETR and Swin-UNETR, 3D UX-Net and MA3DSeg segmentation results show that the accuracy of unsmooth tumor boundary pixels classification is improved, and the final result is more close to the label image. MA3DSeg effectively improve endometrial cancer edge region mis-segmentation with high segmentation accuracy. Compared with the competing networks, when segmenting discontinuous and tiny tumors in fifth group images, MA3DSeg has a better performance in dealing with the edges than ISA-Net and MEFN, effectively segmenting complex morphology tumors and suppressing background noise.

Cross-Validation Experiment

Due to the scattered lymphomas all over the body and complex morphology, it is challenging for accurate segmentation. Thus we chose AutoPET dataset for cross-validation experiments to validate the generalization ability of MA3DSeg. As shown in Table 3, the Dice achieves the optimal value of 0.723 in MA3DSeg, which is 1.5% higher than the sub-optimal 3DUX-Net. The RVD is 1.2% lower than the optimal value, which indicates

	ECPC-IDS				Hecktor 2022			
method	Dice	RVD	HD95	ASSD	Dice	RVD	HD95	ASSD
VNet	0.732	0.356	11.486	2.868	0.771	0.435	8.165	2.473
nnUNet	0.746	0.332	11.232	2.986	0.786	0.322	7.558	2.335
AttnUNet	0.728	0.344	10.876	2.657	0.768	0.318	7.132	2.186
UNETR	0.748	0.318	10.665	2.517	0.793	0.377	6.878	1.985
Swin-UNETR	0.751	0.297	9.487	2.704	0.804	0.354	6.559	2.065
3D UX-Net	0.774	0.288	10.213	2.531	0.807	0.306	5.834	1.976
ISA-Net	0.779	0.291	9.756	2.664	0.788	0.314	5.775	1.874
MEFN	0.781	0.273	9.332	2.547	0.810	0.323	5.322	1.869
ours	0.783	0.267	9.878	2.443	0.812	0.321	4.887	1.884

Table 2. Quantitative comparison results on ECPC-IDS and Hecktor 2022 datasets.

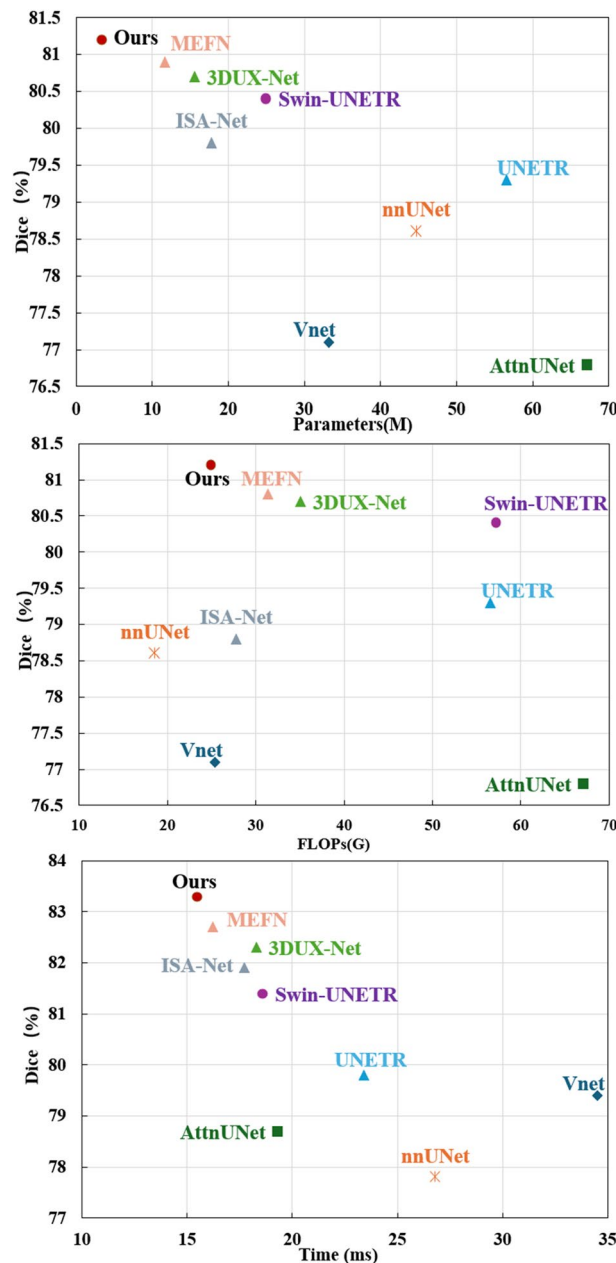


Figure 5. The algorithm performance of the proposed network is compared with the competitive network..

that there is some error in the segmentation volume and the true volume. Moreover, our proposed network reaches optimal values in HD95 and ASSD metrics. Therefore, cross-validation experiment results show that MA3DSeg achieves strong generalization ability for 3D tumor segmentation compared to competing methods.

The visualisation of segmentation effect on AutoPET dataset is shown as Fig.7. In the first and second groups, AttnUNet and Swin-UNETR can accurately localize dispersed tumors, but segmentation of tiny tumors is seen to misclassify the target as normal organ tissues. In the first group, when segmenting small lesions ($< 5mm^3$) in target tumors, there is a decrease in various metrics compared to all samples. Dice drops from 0.723 to 0.675, and ASSD decreases from 2.824 to 3.017, with reductions of 7.1% and 6.8% respectively. Compared with the competing networks, MA3DSeg performs well in the task of segmenting complex edges and junctions of 3D volumetric medical images as shown in Fig.8.

The above-mentioned quantitative and qualitative results show that compared with nnUNet, MA3DSeg achieves the best performance in Dice on the ECPC-IDS, Hecktor 2022, and AutoPET datasets, with improvements of 3.7%, 2.6%, and 4.1% respectively. This indicates that encoder-decoder based methods are effective, but other methods have limitations in long-range context fusion, which may lose detailed information such as edges and textures. UNETR and Swin-UNETR use pure Transformer as the encoder, but their results on the AutoPET dataset are not satisfactory. When compared with ISA-Net and MEFN, our proposed 3D-DX block

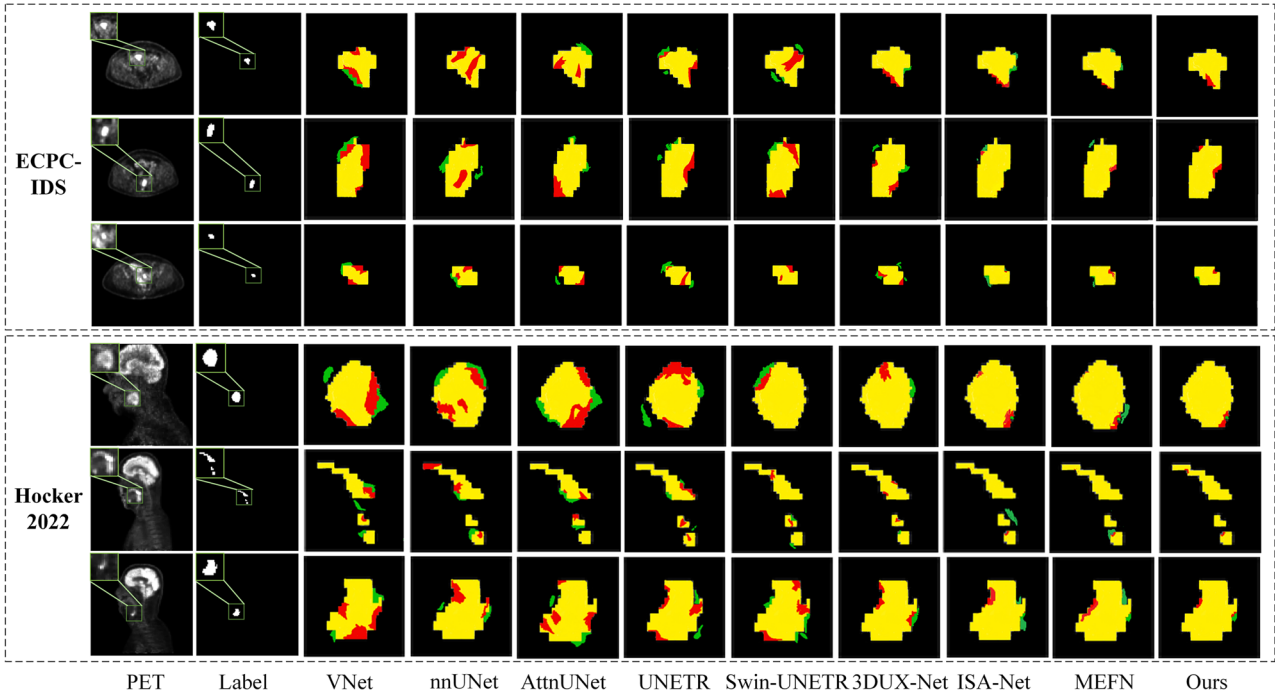


Figure 6. Qualitative comparison of the sagittal plane of images on the ECPC-IDS and Hocker datasets (The yellow, green, and red colors in each prediction map represent the true positive, false positive, and false negative regions, respectively).

Method	Dice	RVD	HD95	ASSD
VNet	0.678	0.417	15.265	3.251
nnUNet	0.682	0.425	14.552	3.208
AttnUNet	0.669	0.411	14.189	3.056
UNETR	0.695	0.396	13.865	2.997
Swin-UNETR	0.714	0.385	13.118	2.859
3D UX-Net	0.708	0.391	12.568	2.914
ISA-Net	0.711	0.387	11.441	2.907
MEFN	0.708	0.316	11.087	2.887
ours	0.723	0.328	10.445	2.824

Table 3. The results of cross-validation experiments with competitive methods on AutoPET dataset.

based on hierarchical transformers with s also demonstrates better performance in capturing local details and providing coarse features for subsequent detail recovery.

Ablation Study

We perform ablation experiments to validate the contribution of each module to the network as shown in Table 4.

Effectiveness of 3D-DX Block

After adding the 3D-DX block to the baseline, the segmentation metrics were improved on all three datasets. Dice is improved by 0.7%, 1.3% and 1.7% on ECPC-IDS , Hecktor 2022 and AutoPET, respectively. RVD, HD95 and ASSD scores are improved to 0.306, 13.589 and 3.114 on Hecktor 2022 dataset. This suggests that the 3D-DX block plays a role in capturing a large receptive field, resulting in effective global contextual information that improves the accuracy of classification of segmented 3D voxels.

Effectiveness of MAFE

Ater adding MAFE module, the Dice score is improved from 0.741 to 0.792 on AutoPET dataset. The RVD, HD95, and ASSD are improved by 0.004, 1.174, and 0.228 compared to without the MAFE module. Therefore, the MAFE module strengthens the 3D images self attention mechanism and enhances the expressive ability between spatial and channel features, which helps to achieve beter tumor segmentation as shown in Fig.9.

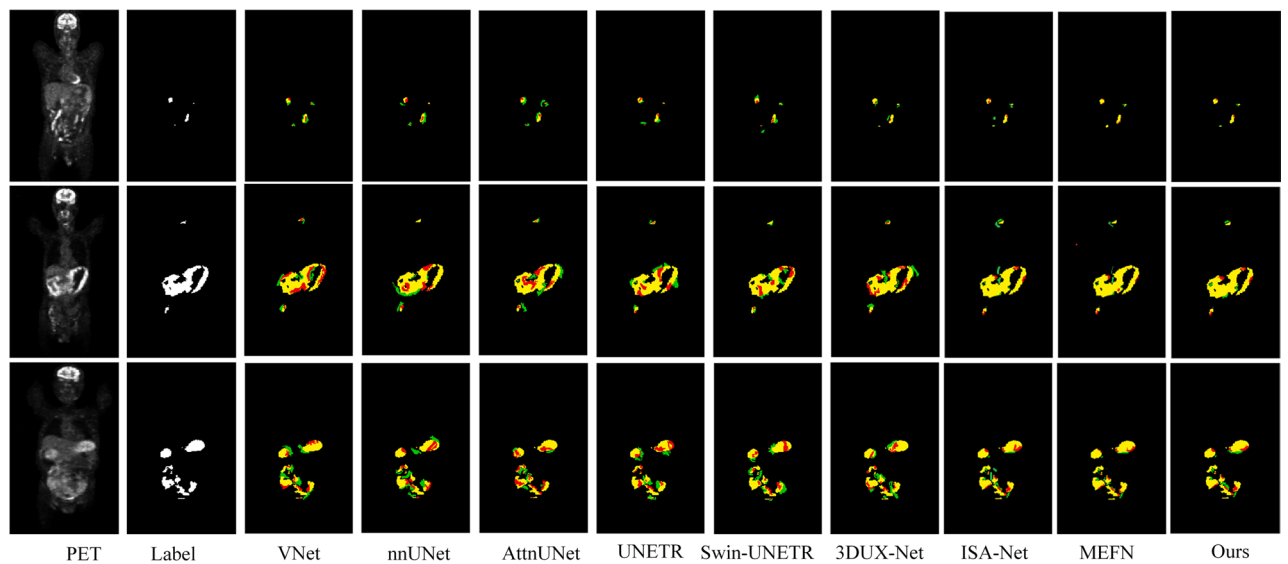


Figure 7. Comparison of results between different methods on AutoPET dataset.

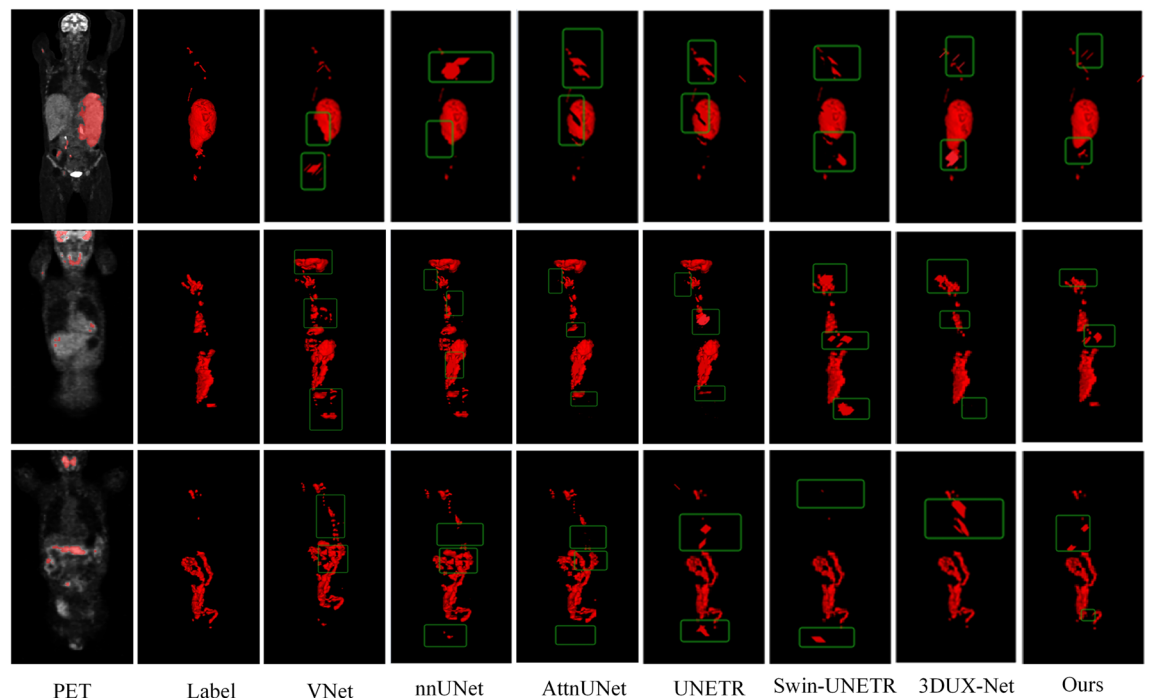


Figure 8. 3D segmentation visualisation results on AutoPET dataset.

Effectiveness of IBS

When the IBS module is integrated alone into the baseline model, Dice, RVD, HD95, and ASSD are improved by 3.7%, 7.7%, 11.7%, and 12.2% respectively on ECPC-IDS. The results demonstrate that the IBS module independently contributes to the tumor boundary segmentation.

Qualitative visualization of the segmentation results of the IBS module as shown in Fig.10, where ∇F : Raw differential features. $\cdot$: Deep features from the precedent decoder layer. $g(|\nabla P|^2)$: Semantic guidance. $g(|\nabla P|^2) \times \nabla P$: Highlighted boundary features. $g(|\nabla P|^2) \times \nabla P + F$: Enhanced skipped features.

Method				ECPC-IDS				Hecktor 2022				AutoPET			
Baseline	3D-DX	MA-FE	IBS	Dice	RVD	HD95	ASSD	Dice	RVD	HD95	ASSD	Dice	RVD	HD95	ASSD
✓	–	–	–	0.685	0.332	13.528	3.452	0.713	0.317	14.552	3.258	0.724	0.298	12.008	2.956
✓	✓	–	–	0.692	0.315	12.274	3.110	0.726	0.306	13.589	3.114	0.741	0.285	11.526	2.885
✓	–	–	✓	0.711	0.308	12.106	3.074	0.744	0.296	12.263	2.902	0.767	0.279	10.843	2.754
✓	✓	✓	–	0.714	0.301	12.621	3.118	0.765	0.294	11.254	3.065	0.792	0.281	10.352	2.657
✓	–	✓	✓	0.708	0.314	12.055	3.057	0.738	0.284	12.115	2.881	0.785	0.276	11.854	2.714
✓	✓	–	✓	0.703	0.305	12.286	3.162	0.755	0.304	11.528	2.957	0.817	0.294	11.557	2.763
✓	✓	✓	✓	0.735	0.296	11.553	2.986	0.767	0.273	10.424	2.884	0.832	0.273	10.122	2.548

Table 4. Comparison table of segmentation results for ablation experiments of our model.

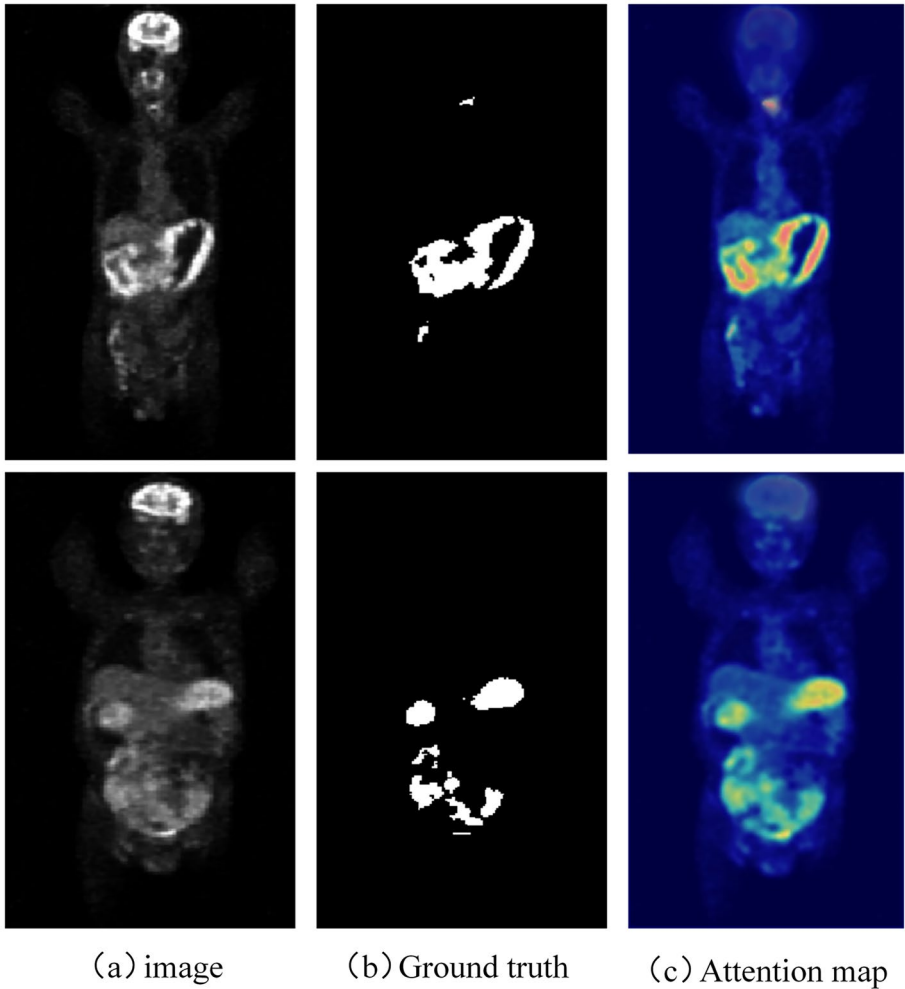


Figure 9. Visualization results of MAFE module on AutoPET dataset. For each row, we show an input image and corresponding ground truth, and we give a multiscale attention-guided map.

Loss Function

In order to speed up convergence during the training phase, we used a loss function that contains a linear combination of the Focal loss function and the Dice loss functions. MA3DSeg is trained with two loss functions on AutoPET dataset. In Table 5, the Dice and ASSD scores are improved by 0.025 and 0.488 respectively.

Conclusion

In this paper, we present a lightweight MA3DSeg network for segmenting 3D tumors of PET images. We innovatively proposed 3D-DX block for the backbone network. Specifically, using the ConvNet block to adjust the inherent properties of the Swin Transformer to improve the volumetric segmentation performance with a smaller model capacity. MAFE is designed to perform feature enhancement from the interaction between spatial

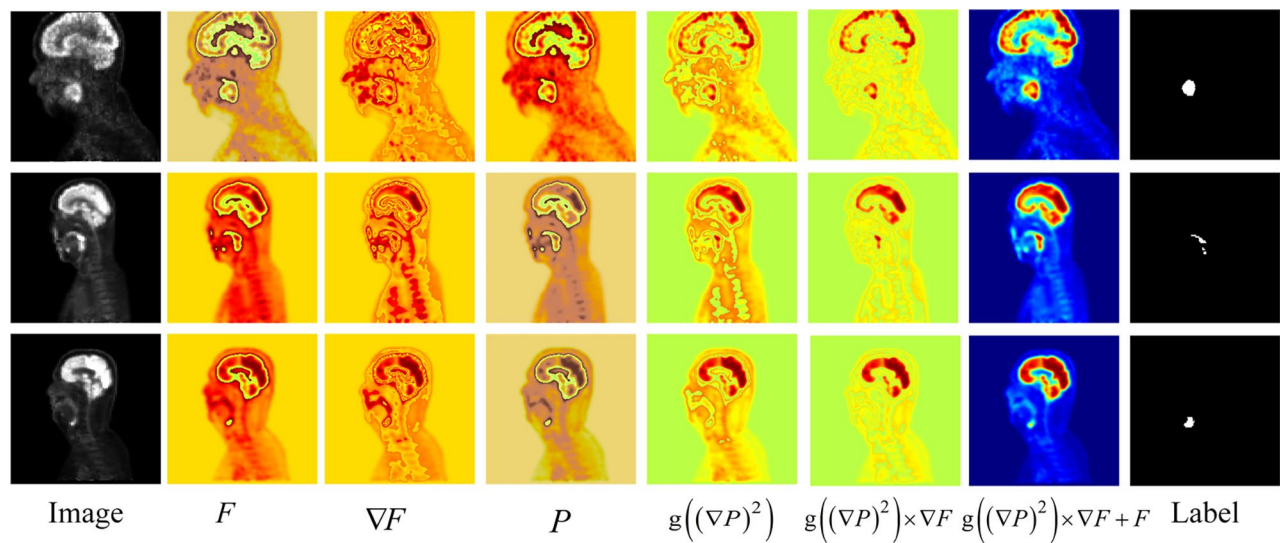


Figure 10. Qualitative visualization of the segmentation results of the IBS module on Hocker 2022, improving the performance of segmentation edges..

Method	Dice	RVD	HD95	ASSD
MA3DSeg+ L_{Dice}	0.756	0.325	12.556	3.331
MA3DSeg+ $L_{Dice}+L_{Focal}$	0.781	0.308	10.254	2.843

Table 5. Ablation study of the loss function ($\alpha \in [0.3, 0.7], \beta = 1 - \alpha$).

and channel features to effectively suppress the interference of background noise. In addition, IBS module utilises explicit and implicit difference information are greatly improve the representation of inter-class boundaries and enhance boundary features. MA3DSeg is conducted sufficient experiments on three datasets, the results show that MA3DSeg achieves the optimal performance in 3D tumor segmentation compared to competing networks. MA3DSeg has great potential for clinical application in medical image segmentation.

Data Availability

In this article, we used three publicly available dataset. The ECPC-IDS dataset can be found in <https://figshare.com/articles/dataset/ECPC-IDS/23808258>. The Hecktor 2022 dataset can be found in <https://hecktor.grand-challenge.org/Data/>. The AutoPET dataset is a publicly available dataset of the 25th MICCAI challenge. Source codes are available at <https://github.com/Lincen811/MA3DSeg>.

Received: 23 April 2025; Accepted: 17 October 2025
Published online: 07 November 2025

References

1. Kubota, K. From tumor biology to clinical PET: a review of positron emission tomography (PET) in oncology. *Ann. Nucl. Med.* **15**, 471–486 (2001).
2. Whybra, P., Parkinson, C., Foley, K., Staffurth, J. & Spezi, E. Assessing radiomic feature robustness to interpolation in 18F-FDG PET imaging. *Sci. Rep.* **9**(1), 9649 (2019).
3. Hatamizadeh, A., Tang, Y., Nath, V., Yang, D., Myronenko, A., Landman, B., et al. Unetr: Transformers for 3d medical image segmentation. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision; (2022). 574–584.
4. Hatamizadeh, A., Nath, V., Tang, Y., Yang, D., Roth, H.R., Xu, D. Swin unetr: Swin transformers for semantic segmentation of brain tumors in mri images. In: International MICCAI brainlesion workshop; 272–284 (2021). Springer.
5. Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., et al. Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV); 10012–10022 (2021).
6. Zhang, J., Xie, Y., Wang, Y. & Xia, Y. Inter-slice context residual learning for 3d medical image segmentation. *IEEE Trans. Med. Imaging* **2**, 40 (2021).
7. Zhou, Z., Rahman Siddiquee, M.M., Tajbakhsh, N., Liang, J. Unet++: A nested u-net architecture for medical image segmentation. In: Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: 4th International Workshop; 3–11 (2018). Springer.
8. Kamnitsas, K. et al. Efficient multi-scale 3D CNN with fully connected CRF for accurate brain lesion segmentation. *Med. Image Anal.* **36**, 61–78 (2017).
9. Isensee, F., Jaeger, P. F., Kohl, S. A., Petersen, J. & Maier-Hein, K. H. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nat. Methods* **18**(2), 203–211 (2021).
10. Jaderberg, M., Simonyan, K., Zisserman, A., et al. Spatial transformer networks. *Advances in neural information processing systems*. 28 (2015).

11. Hu J, Shen L, Sun G. Squeeze-and-excitation networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2018. pp. 7132–7141.
12. Peng, H., Chen, X., Zhao, J. Residual pixel attention network for spectral reconstruction from RGB images. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops; 486–487 (2020).
13. Xie, X. et al. PIF-Net: A parallel interweave fusion network for knee joint segmentation. *Biomed. Signal Process. Control* **109**, 107967 (2025).
14. Xie, X. et al. Discriminative features pyramid network for medical image segmentation. *Biocybernet. Biomed. Eng.* **44**(2), 327–340 (2024).
15. Gerard, S. E. et al. Multi-resolution convolutional neural networks for fully automated segmentation of acutely injured lungs in multiple species. *Med. Image Anal.* **60**, 101592 (2020).
16. Xie, X. et al. A context hierarchical integrated network for medical image segmentation. *Comput. Electr. Eng.* **101**, 108029 (2022).
17. Fu, H. et al. HMRNet: High and multi-resolution network with bidirectional feature calibration for brain structure segmentation in radiotherapy. *IEEE J. Biomed. Health Inform.* **26**(9), 4519–4529 (2022).
18. Lee, HH., Bao, S., Huo, Y., Landman, BA. 3D UX-Net: A Large Kernel Volumetric ConvNet Modernizing Hierarchical Transformer for Medical Image Segmentation. arXiv preprint [arXiv:2209.15076](https://arxiv.org/abs/2209.15076). (2022).
19. Zou, Z., Zou, B., Kui, X., Chen, Z. & Li, Y. DGCBG-Net: A dual-branch network with global cross-modal interaction and boundary guidance for tumor segmentation in PET/CT images. *Comput. Methods Programs Biomed.* **250**, 108125 (2024).
20. Liu, L., Hu, Z., Dai, Y., Ma, X. & Deng, P. ISA: Ingenious Siamese attention for object detection algorithms towards complex scenes. *ISA Trans.* **143**, 205–220 (2023).
21. Cheng, T., Wang, X., Huang, L., Liu, W. Boundary-preserving mask r-cnn. In: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XIV 16; 660–676 (2020). Springer.
22. Acuna, D., Kar, A., Fidler, S. Devil is in the edges: Learning semantic boundaries from noisy annotations. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 11075–11083 (2019).
23. Chen, Z. et al. ALIEN: Attention-guided cross-resolution collaborative network for 3D gastric cancer segmentation in CT images. *Biomed. Signal Process. Control* **96**, 106500 (2024).
24. Wu, Y., Jiang, H. & Pang, W. MSRA-Net: Tumor segmentation network based on multi-scale residual attention. *Comput. Biol. Med.* **158**, 106818 (2023).
25. Oktay, O., Schlemper, J., Folgoc, LL., Lee, M., Heinrich, M., Misawa, K., et al. Attention u-net: Learning where to look for the pancreas. arXiv preprint [arXiv:1804.03999](https://arxiv.org/abs/1804.03999).10 (2018).
26. Xie, X., Zhang, W. & Pan, X. Canet: Context aware network with dual-stream pyramid for medical image segmentation. *Biomed. Signal Process. Control* **81**, 104437 (2023).
27. Kang, S. et al. Learning feature fusion via an interpretation method for tumor segmentation on PET/CT. *Appl. Soft Comput.* **148**, 110825 (2023).
28. Niu, X., Lian, J., Zhang, H., Zhang, C., Dong, Z. A lung cancer tumor image segmentation method of a SFC-MSPCNN based on PET/CT. In: 2021 International Conference on Computer, Internet of Things and Control Engineering (CITCE); 69–73 (2021). IEEE.
29. Diao, Z., Jiang, H., Han, X. H., Yao, Y. D. & Shi, T. EFNet: Evidence fusion network for tumor segmentation from PET-CT volumes. *Phys. Med. Biol.* **66**(20), 205005 (2021).
30. Liu, Z., Mao, H., Wu, CY., Feichtenhofer, C., Darrell, T., Xie, S. A convnet for the 2020s. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 11976–11986 (2022).
31. Sapiro, G. *Geometric partial differential equations and image analysis* (Cambridge University Press, 2006).
32. Pavl, S. D. & Craus, M. Reaction-diffusion model applied to enhancing U-Net accuracy for semantic image segmentation. *Discr. Cont. Dyn. Syst. -S.* **16**(1), 54–74 (2023).
33. Tang, D. et al. ECPC-IDS: A benchmark endometrial cancer PET/CT image dataset for evaluation of semantic segmentation and detection of hypermetabolic regions. *Comput. Biol. Med.* **171**, 108217 (2024).
34. Oreiller, V. et al. Head and neck tumor segmentation in PET/CT: the HECKTOR challenge. *Med. Image Anal.* **77**, 102336 (2022).
35. Huang, Zhengyong et al. ISA-Net: Improved spatial attention network for PET-CT tumor segmentation. *Comput. Methods Programs Biomed.* **226**, 107129 (2022).
36. Qi, Yuxuan et al. Multi-modal evidential fusion network for trustworthy PET/CT tumor segmentation. *Knowl.-Based Syst.* **324**, 113838 (2025).

Author contributions

LJ: data preprocessing and model construction. XZ: original draft preparation. WX: writing-reviewing. All authors read and approved the final manuscript.

Funding

This work was supported in Future Network Scientific Research Fund Project (No. FNSRFP-2021-YB-15), Jiangsu Key Development Planning Project (BE2023004-2), General Program of the Nanjing Drum Tower Hospital Clinical Research (2024-LCYJ-MS-25), Natural Science Foundation of Jiangsu Province (Higher Education Institutions) (20KJA520001), China Postdoctoral Science Foundation (2024M762244), Cultivation Program of the Nanjing Drum Tower Hospital Clinical Research (2022-LCYJ-PY-44).

Jiangsu Key Development Planning Project (BE2023004-2), General Program of the Nanjing Drum Tower Hospital Clinical Research (2024-LCYJ-MS-25), Natural Science Foundation of Jiangsu Province (Higher Education Institutions) (20KJA520001), China Postdoctoral Science Foundation (2024M762244), Cultivation Program of the Nanjing Drum Tower Hospital Clinical Research (2022-LCYJ-PY-44).

Declarations

Competing interests

The authors declare no competing interests.

Consent to Participate

This study did not involve human subjects.

Consent to Publish

This study has no individual person data.

Additional information

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